

MongoDB Integration for CISD 43 Final Using the Creepypastas Dataset

In [2]: *#first, one must start the MongoDB session in Python*

```
from pymongo import MongoClient

#import pandas to allow dataset manipulation

import pandas as pd
```

In [3]: *#now, one must set up the Mongo client*

```
client = MongoClient('mongodb://localhost:27017/')
```

In [4]: *#one must select the database creepypasta_db and the collection stories*

```
db = client['creepypasta_db']
collection = db['stories']
```

In [5]: *#now, the first three records will be retrieved and printed (truncate the long body*

```
shortened_retrieval = collection.find().limit(3)

for doc in shortened_retrieval:
    body = doc.get('body', '')
    preview = body[:200] + '...' if len(body) > 200 else body

    print({
        'story_name': doc.get('story_name'),
        'estimated_reading_time': doc.get('estimated_reading_time'),
        'publish_date': doc.get('publish_date'),
        'average_rating': doc.get('average_rating'),
        'tags': doc.get('tags'),
        'categories': doc.get('categories'),
        'body_preview': preview
    })
```

```
{'story_name': 'If You're Armed and at the Glenmont Metro, Please Shoot Me', 'estimated_reading_time': '12 minutes', 'publish_date': 'February 20, 2020 ', 'average_rating': 9.25, 'tags': ' drug trials, drugs, experimentation, experiments, madness, Peter Frost David, psychological, psychological horror, sci-fi, science, science fiction, time\n', 'categories': ' Drugs and Addictions, Madness, Paranoia, and Mental Illness, Psychological Horror, Science and Experimentation, Science Fiction and Aliens ', 'body_preview': 'If you're armed and at the Glenmont metro, please shoot me.\nMake it a headshot. Shoot me in the temple, aiming slightly downwards. I need the bullet to travel the shortest possible distance through my...'}
{'story_name': 'A Shattered Life', 'estimated_reading_time': '14 minutes', 'publish_date': 'July 22, 2020 ', 'average_rating': 9.13, 'tags': ' disease, entities, Matt Dymerski, monsters, paranormal, supernatural, time travel\n', 'categories': ' Madness, Paranoia, and Mental Illness, Monsters, Creatures, and Cryptids, Science Fiction and Aliens ', 'body_preview': ' I don't know when you're going to read this, but I can tell you when it started: I was out for a walk alone in the woods when the entity came for me. It was beyond a blur. It was, for lack of a better...'}
{'story_name': 'My Father Punished Me When I Talked to Ghosts', 'estimated_reading_time': '8 minutes', 'publish_date': 'December 10, 2019 ', 'average_rating': 9.12, 'tags': ' abductions, blindness, child abuse, childhood, children, crimes, deaths, deception, investigations, kidnappings, kids, lies, murders\n', 'categories': ' Abductions and Kidnappings, Children and Childhood, Deaths, Murders, and Disappearances, Investigations and Crimes ', 'body_preview': 'I've been blind since birth. As I grew up, everything was described to me in such vivid detail that I didn't even realize why it was that important to see, especially having no reference point to compare...'}

```

In [6]: *#now one will find the story with the title - The Whistlers: Ruth's Account*

```
whistlers = collection.find_one({'story_name': "The Whistlers: Ruth's Account"})
preview = body[:200] + '...' if len(body) > 200 else body

print({
    'story_name': whistlers.get('story_name'),
    'estimated_reading_time': whistlers.get('estimated_reading_time'),
    'publish_date': whistlers.get('publish_date'),
    'average_rating': whistlers.get('average_rating'),
    'tags': whistlers.get('tags'),
    'categories': whistlers.get('categories'),
    'body_preview': preview
})

```

```
{'story_name': 'The Whistlers: Ruth's Account', 'estimated_reading_time': '54 minutes', 'publish_date': 'July 8, 2019 ', 'average_rating': 9.05, 'tags': ' Amity Argot, beings, creatures, diaries, entities, forests, journals, nature, outdoors, strange, The Whistlers Series, trees, unexplained, woods\nPlease wait...\n', 'categories': ' Beings and Entities, Journals and Diaries, Nature and the Outdoors, Strange and Unexplained ', 'body_preview': 'I've been blind since birth. As I grew up, everything was described to me in such vivid detail that I didn't even realize why it was that important to see, especially having no reference point to compare...'}

```

In [7]: *#now, one will find the total amount of documents*

```
total_docs = collection.count_documents({})
print('total documents:', total_docs)

```

total documents: 3510

In [9]: *#now, find all stories with a rating above 9*

```
above_eight Rated = collection.find({
    'average_rating': {'$gt': 9}},
    {'story_name': 1,
     'average_rating': 1,
     '_id': 0
})

for doc in above_eight Rated:
    print(doc)
```

```
{'story_name': 'If You're Armed and at the Glenmont Metro, Please Shoot Me', 'average_rating': 9.25}
{'story_name': 'A Shattered Life', 'average_rating': 9.13}
{'story_name': 'My Father Punished Me When I Talked to Ghosts', 'average_rating': 9.12}
{'story_name': 'Distorted Warning Signals', 'average_rating': 9.12}
{'story_name': 'He Who Wanders', 'average_rating': 9.12}
{'story_name': 'The Living History Project', 'average_rating': 9.12}
{'story_name': 'I Found a Letter From My Stalker', 'average_rating': 9.12}
{'story_name': 'Don't Let Them In', 'average_rating': 9.11}
{'story_name': 'What Happens When the Stars Go Out', 'average_rating': 9.11}
{'story_name': 'The Seer of Possibilities', 'average_rating': 9.11}
{'story_name': 'Psychosis', 'average_rating': 9.11}
{'story_name': 'Acts of Vengeance', 'average_rating': 9.1}
{'story_name': 'Mommy Sleeps in the Basement', 'average_rating': 9.1}
{'story_name': 'Charles Bonnet Syndrome', 'average_rating': 9.1}
{'story_name': 'The Well', 'average_rating': 9.1}
{'story_name': 'Sniff', 'average_rating': 9.1}
{'story_name': 'Turn It Off', 'average_rating': 9.1}
{'story_name': 'Bound', 'average_rating': 9.1}
{'story_name': 'The Melancholy of Herbert Solomon', 'average_rating': 9.1}
{'story_name': 'Bedtime', 'average_rating': 9.1}
{'story_name': 'Sisters', 'average_rating': 9.09}
{'story_name': 'Ticci-Toby', 'average_rating': 9.09}
{'story_name': 'I Drive for Cerber', 'average_rating': 9.09}
{'story_name': 'The Quiet Sky', 'average_rating': 9.09}
{'story_name': 'Forget Me', 'average_rating': 9.09}
{'story_name': 'The Fairies', 'average_rating': 9.09}
{'story_name': 'Ascension', 'average_rating': 9.09}
{'story_name': 'Slum', 'average_rating': 9.09}
{'story_name': 'My Grandfather's Final Invention', 'average_rating': 9.09}
{'story_name': 'Three Friends Diner', 'average_rating': 9.09}
{'story_name': 'The Face of Fear', 'average_rating': 9.09}
{'story_name': 'In the Walls', 'average_rating': 9.09}
{'story_name': 'Artificial', 'average_rating': 9.09}
{'story_name': 'The Devil Game', 'average_rating': 9.09}
{'story_name': 'On a Hill', 'average_rating': 9.09}
{'story_name': 'Mr. Widemouth', 'average_rating': 9.09}
{'story_name': 'The Art Of Jacob Emory', 'average_rating': 9.09}
{'story_name': 'The Gift Of Mercy', 'average_rating': 9.09}
{'story_name': 'Dust', 'average_rating': 9.09}
{'story_name': 'The Promise', 'average_rating': 9.08}
{'story_name': 'The Whistlers: Bill's Account', 'average_rating': 9.08}
{'story_name': 'Dandelions', 'average_rating': 9.08}
{'story_name': 'The Last Man of Faith', 'average_rating': 9.08}
{'story_name': 'Blindness', 'average_rating': 9.08}
{'story_name': 'The Lost Cosmonaut', 'average_rating': 9.08}
{'story_name': 'The Perfectly Behaved Boy', 'average_rating': 9.08}
{'story_name': 'The Cats of Juniper Valley', 'average_rating': 9.08}
{'story_name': 'The Fourth Child', 'average_rating': 9.08}
{'story_name': 'Promises', 'average_rating': 9.08}
{'story_name': 'Sadie', 'average_rating': 9.08}
{'story_name': 'The Sandman', 'average_rating': 9.08}
{'story_name': 'The Systelien Specter', 'average_rating': 9.08}
{'story_name': 'Pretend', 'average_rating': 9.08}
{'story_name': 'The Tulpa', 'average_rating': 9.08}
```



```
{'story_name': 'Lightning', 'average_rating': 9.08}
{'story_name': 'An Egg', 'average_rating': 9.08}
{'story_name': 'Survival Guide', 'average_rating': 9.08}
{'story_name': 'The Body Farm', 'average_rating': 9.07}
{'story_name': 'The Twitch Stream', 'average_rating': 9.07}
{'story_name': 'Ted the Caver', 'average_rating': 9.07}
{'story_name': 'When Science Found God', 'average_rating': 9.07}
{'story_name': 'Magda', 'average_rating': 9.07}
{'story_name': 'The Blaganschlor', 'average_rating': 9.07}
{'story_name': 'Maisie Went Missing Last Year', 'average_rating': 9.07}
{'story_name': 'The Pastel Man', 'average_rating': 9.07}
{'story_name': 'The Deafening', 'average_rating': 9.07}
{'story_name': 'Seventeen', 'average_rating': 9.07}
{'story_name': 'Forgotten Valentine', 'average_rating': 9.07}
{'story_name': 'In The Land of Black and White', 'average_rating': 9.07}
{'story_name': 'Dear Abby', 'average_rating': 9.07}
{'story_name': 'Ickbarr Bigelsteine', 'average_rating': 9.07}
{'story_name': 'The Russian Sleep Experiment', 'average_rating': 9.07}
{'story_name': 'Midnight in Texas', 'average_rating': 9.06}
{'story_name': 'A Peculiar Kind of Madness', 'average_rating': 9.06}
{'story_name': 'Secret Bar', 'average_rating': 9.06}
{'story_name': 'The Killer's Interview', 'average_rating': 9.06}
{'story_name': 'The Final Fate of Mary Ford', 'average_rating': 9.06}
{'story_name': 'When Gods Blink', 'average_rating': 9.06}
{'story_name': 'The Skinwalker', 'average_rating': 9.06}
{'story_name': 'The Crawlspace', 'average_rating': 9.06}
{'story_name': 'Julia Legare', 'average_rating': 9.06}
{'story_name': 'My Wives Don't Get Along', 'average_rating': 9.05}
{'story_name': 'The Girl the Universe Forgot', 'average_rating': 9.05}
{'story_name': 'Daughter's Drawings', 'average_rating': 9.05}
{'story_name': 'The Whistlers: Ruth's Account', 'average_rating': 9.05}
{'story_name': 'Playpen', 'average_rating': 9.05}
{'story_name': 'The Reaper's Scythe', 'average_rating': 9.05}
{'story_name': 'Rose - Part 1', 'average_rating': 9.05}
{'story_name': 'Family of Three Plus One', 'average_rating': 9.05}
{'story_name': 'Crying Numbers', 'average_rating': 9.05}
{'story_name': 'Too Long at the Cliff', 'average_rating': 9.05}
{'story_name': 'Bright Eyes', 'average_rating': 9.05}
{'story_name': 'The Pursuit Institute', 'average_rating': 9.05}
{'story_name': 'Mary', 'average_rating': 9.05}
{'story_name': 'A Christmas Feast', 'average_rating': 9.05}
{'story_name': 'Face Your Fears', 'average_rating': 9.05}
{'story_name': 'I Am the Apocalypse', 'average_rating': 9.05}
{'story_name': 'Fenter Woods', 'average_rating': 9.05}
{'story_name': 'Something Happened During a Hostage Negotiation I Can't Explain', 'average_rating': 9.04}
{'story_name': 'A Story From the BlackGuard', 'average_rating': 9.04}
{'story_name': 'Tales From the Gas Station', 'average_rating': 9.04}
{'story_name': 'Scorpion River', 'average_rating': 9.04}
{'story_name': 'Willow Creek', 'average_rating': 9.04}
{'story_name': '12 Steps', 'average_rating': 9.04}
{'story_name': 'Three Truths', 'average_rating': 9.04}
{'story_name': 'The Tollman', 'average_rating': 9.04}
{'story_name': 'Leo', 'average_rating': 9.04}
{'story_name': 'Ubloo', 'average_rating': 9.04}
{'story_name': 'It Has No Face', 'average_rating': 9.04}
```

```
{'story_name': 'Wendall Lane Diaries: You Shouldn't', 'average_rating': 9.04}
{'story_name': 'Rabbits in the Creek', 'average_rating': 9.04}
{'story_name': 'The Door', 'average_rating': 9.04}
{'story_name': 'String Theory', 'average_rating': 9.04}
{'story_name': 'Doors', 'average_rating': 9.04}
{'story_name': 'Candle Cove', 'average_rating': 9.04}
{'story_name': 'Persuaded', 'average_rating': 9.04}
{'story_name': 'World's Best School Psychologist', 'average_rating': 9.03}
{'story_name': 'Dawn in Texas', 'average_rating': 9.03}
{'story_name': 'My Son Committed Suicide, and My Wife Blames Me', 'average_rating':
9.03}
{'story_name': 'Restoration', 'average_rating': 9.03}
{'story_name': 'An Open Letter To My Daughter's Killer', 'average_rating': 9.03}
{'story_name': 'Human Nature', 'average_rating': 9.03}
{'story_name': 'Ben Drowned', 'average_rating': 9.03}
{'story_name': 'Barter', 'average_rating': 9.03}
{'story_name': 'Razor Games', 'average_rating': 9.03}
{'story_name': 'Smile Dog', 'average_rating': 9.03}
{'story_name': 'The Only Sensible Ritual Pasta', 'average_rating': 9.03}
{'story_name': 'Sarah O'Bannon', 'average_rating': 9.03}
{'story_name': 'Knocking', 'average_rating': 9.02}
{'story_name': 'I Am Halloween', 'average_rating': 9.02}
{'story_name': 'The Magic Show From Hell', 'average_rating': 9.02}
{'story_name': 'The Strange Case of Edmonson, Kentucky', 'average_rating': 9.02}
{'story_name': 'Devil in the Details', 'average_rating': 9.02}
{'story_name': 'The Three Day Dark', 'average_rating': 9.02}
{'story_name': 'The Space Above the Closet', 'average_rating': 9.02}
{'story_name': 'Mr. Leaves', 'average_rating': 9.02}
{'story_name': 'Star Light', 'average_rating': 9.02}
{'story_name': 'Nearby', 'average_rating': 9.02}
{'story_name': 'Never Again', 'average_rating': 9.02}
{'story_name': 'My Fear Of Water', 'average_rating': 9.02}
{'story_name': 'Beyond Vantablack', 'average_rating': 9.01}
{'story_name': 'All of Our Mistakes Are Never Forgotten', 'average_rating': 9.01}
{'story_name': 'Carlisle Pond', 'average_rating': 9.01}
{'story_name': 'Night In A Tree Stand', 'average_rating': 9.01}
{'story_name': 'My Creation', 'average_rating': 9.01}
{'story_name': 'I Found a Dead Girl's Diary', 'average_rating': 9.01}
{'story_name': 'You're Not Afraid', 'average_rating': 9.01}
{'story_name': 'The Grimes Home', 'average_rating': 9.01}
{'story_name': 'The Story of Her Holding an Orange: Part One', 'average_rating': 9.0
1}
{'story_name': 'The Scuttler', 'average_rating': 9.01}
```

In [10]: *#find all stories with the word 'night' in their title (make it case insensitive)*

```
stories_with_the_word_night = collection.find({
    'story_name': {'$regex': 'night', '$options': 'i'}},
    {'story_name': 1,
     'average_rating': 1,
     'estimated_reading_time': 1,
     '_id': 0
    })
for doc in stories_with_the_word_night:
    print(doc)
```

```
{'story_name': 'Midnight in Texas', 'average_rating': 9.06, 'estimated_reading_time': '28 minutes'}
{'story_name': 'Night In A Tree Stand', 'average_rating': 9.01, 'estimated_reading_time': '22 minutes'}
{'story_name': 'Strange Things Happen Overnight at Lowe's', 'average_rating': 8.96, 'estimated_reading_time': '52 minutes'}
{'story_name': 'The Midnight Lock-Up', 'average_rating': 8.85, 'estimated_reading_time': '24 minutes'}
{'story_name': 'Never Answer the Door at Night', 'average_rating': 8.82, 'estimated_reading_time': '4 minutes'}
{'story_name': 'Just Another Midnight', 'average_rating': 8.82, 'estimated_reading_time': '13 minutes'}
{'story_name': 'The Night Springs Cemetery', 'average_rating': 8.71, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Monsters Are Better Than Nightmares', 'average_rating': 8.69, 'estimated_reading_time': '10 minutes'}
{'story_name': 'Will You Sleep Tonight?', 'average_rating': 8.63, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'Something Was in My Kitchen Last Night', 'average_rating': 8.62, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Five Day. Five Nights', 'average_rating': 8.57, 'estimated_reading_time': '6 minutes'}
{'story_name': 'The Night Wire', 'average_rating': 8.53, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Be Home by Midnight on Halloween Night', 'average_rating': 8.51, 'estimated_reading_time': '8 minutes'}
{'story_name': 'The Night Lights', 'average_rating': 8.51, 'estimated_reading_time': '10 minutes'}
{'story_name': 'Things That Go Bump in the Night', 'average_rating': 8.48, 'estimated_reading_time': '13 minutes'}
{'story_name': 'It Comes at Night', 'average_rating': 8.46, 'estimated_reading_time': '7 minutes'}
{'story_name': 'Lazy Saturday Night', 'average_rating': 8.37, 'estimated_reading_time': '2 minutes'}
{'story_name': 'Jamie Lee Curtis and the Night They Filmed Halloween On Our Street', 'average_rating': 8.29, 'estimated_reading_time': '14 minutes'}
{'story_name': 'Something Comes Into My Room Every Night', 'average_rating': 8.17, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Night Shift', 'average_rating': 8.13, 'estimated_reading_time': '8 minutes'}
{'story_name': 'Succession Of Nightmares', 'average_rating': 8.09, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Nightmare Walking', 'average_rating': 8.07, 'estimated_reading_time': '2 minutes'}
{'story_name': 'Night-Time Curiosity', 'average_rating': 8.04, 'estimated_reading_time': '5 minutes'}
{'story_name': 'Late One Fall Night', 'average_rating': 8.04, 'estimated_reading_time': '8 minutes'}
{'story_name': 'Summertime Nightmare', 'average_rating': 8.03, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Good Night / Good Bye', 'average_rating': 8.03, 'estimated_reading_time': '3 minutes'}
{'story_name': 'That Night In The Mirror', 'average_rating': 8.03, 'estimated_reading_time': '4 minutes'}
{'story_name': 'Nightmare', 'average_rating': 8, 'estimated_reading_time': '70 minutes'}
```

```
{'story_name': 'Nightmares', 'average_rating': 7.99, 'estimated_reading_time': '3 minutes'}
{'story_name': 'My Dad Keeps Going Out To The Woods at Night', 'average_rating': 7.96, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Ordinary Nights', 'average_rating': 7.87, 'estimated_reading_time': '6 minutes'}
{'story_name': 'The Night's Hook', 'average_rating': 7.8, 'estimated_reading_time': '10 minutes'}
{'story_name': 'Late Night Television', 'average_rating': 7.8, 'estimated_reading_time': '3 minutes'}
{'story_name': 'Night Terrors', 'average_rating': 7.73, 'estimated_reading_time': '9 minutes'}
{'story_name': 'The Night of the Glass Eyes', 'average_rating': 7.73, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Nightmare', 'average_rating': 7.73, 'estimated_reading_time': '3 minutes'}
{'story_name': 'A Bad Night', 'average_rating': 7.64, 'estimated_reading_time': '9 minutes'}
{'story_name': 'My Friend Became Weirdly Popular Overnight', 'average_rating': 7.59, 'estimated_reading_time': '8 minutes'}
{'story_name': 'Goodnight', 'average_rating': 7.52, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'Night Construction', 'average_rating': 7.52, 'estimated_reading_time': '2 minutes'}
{'story_name': 'The Nightcrawler', 'average_rating': 7.49, 'estimated_reading_time': '8 minutes'}
{'story_name': 'A Horrid Nightmare', 'average_rating': 7.49, 'estimated_reading_time': '8 minutes'}
{'story_name': 'The Cookie Monster at Night', 'average_rating': 7.45, 'estimated_reading_time': '5 minutes'}
{'story_name': 'Nightmare Night', 'average_rating': 7.38, 'estimated_reading_time': '14 minutes'}
{'story_name': 'That Night (Quella Notte)', 'average_rating': 7.36, 'estimated_reading_time': '6 minutes'}
{'story_name': 'The Girl in the Nightgown', 'average_rating': 7.33, 'estimated_reading_time': '13 minutes'}
{'story_name': 'Nightmare', 'average_rating': 7.3, 'estimated_reading_time': '6 minutes'}
{'story_name': 'A Night Terror', 'average_rating': 7.24, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Just Your Average Night', 'average_rating': 7.11, 'estimated_reading_time': '2 minutes'}
{'story_name': 'Waking Nightmares', 'average_rating': 7.08, 'estimated_reading_time': '13 minutes'}
{'story_name': 'The Night of July 13', 'average_rating': 7.08, 'estimated_reading_time': '11 minutes'}
{'story_name': 'Oh, You Know, Just Another Nightmare', 'average_rating': 7.05, 'estimated_reading_time': '4 minutes'}
{'story_name': 'Goodnight', 'average_rating': 7, 'estimated_reading_time': '5 minutes'}
{'story_name': 'Nightmares', 'average_rating': 6.98, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'Where the Nightmares Go', 'average_rating': 6.96, 'estimated_reading_time': '19 minutes'}
{'story_name': 'The Old School Night Nurse', 'average_rating': 6.94, 'estimated_reading_time': '7 minutes'}
```

```
{'story_name': 'They Mostly Come at Night', 'average_rating': 6.84, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Nightmares are Reality', 'average_rating': 6.84, 'estimated_reading_time': '3 minutes'}
{'story_name': 'The First Night of Christmas is for Special People', 'average_rating': 6.83, 'estimated_reading_time': '13 minutes'}
{'story_name': 'Late Night Pizza', 'average_rating': 6.82, 'estimated_reading_time': '4 minutes'}
{'story_name': 'Dear Diary, Goodnight', 'average_rating': 6.82, 'estimated_reading_time': '7 minutes'}
{'story_name': 'Movie Night', 'average_rating': 6.77, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'Goodnight Precious', 'average_rating': 6.7, 'estimated_reading_time': '3 minutes'}
{'story_name': 'Just Your Typical Saturday Night', 'average_rating': 6.65, 'estimated_reading_time': '7 minutes'}
{'story_name': 'The Long Night', 'average_rating': 6.54, 'estimated_reading_time': '7 minutes'}
{'story_name': 'Waking Nightmares', 'average_rating': 6.53, 'estimated_reading_time': '9 minutes'}
{'story_name': 'Night Hike', 'average_rating': 6.48, 'estimated_reading_time': '5 minutes'}
{'story_name': 'Night Nurse', 'average_rating': 6.16, 'estimated_reading_time': '4 minutes'}
{'story_name': 'A Bump in the Night', 'average_rating': 6.04, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Nightmares', 'average_rating': 5.91, 'estimated_reading_time': '6 minutes'}
{'story_name': 'Blue Nightmare', 'average_rating': 5.83, 'estimated_reading_time': '14 minutes'}
{'story_name': 'The Night I Chose To Remember', 'average_rating': 5.79, 'estimated_reading_time': '7 minutes'}
{'story_name': 'Nightmare', 'average_rating': 5.26, 'estimated_reading_time': '4 minutes'}
{'story_name': 'Late Night Metal', 'average_rating': 4.96, 'estimated_reading_time': '7 minutes'}
{'story_name': 'Nightmarish Realities', 'average_rating': 4.59, 'estimated_reading_time': '13 minutes'}
```

```
In [11]: #find the average rating

pipeline = [{'$group': {
    '_id': None,
    'average_ratings_all': {'$avg': '$average_rating'}}}]

avg_rating_all = list(collection.aggregate(pipeline))
print(avg_rating_all)

[{'_id': None, 'average_ratings_all': 7.565045584045585}]
```

```
In [12]: #now, clean the estimated_reading_time column to feature only numbers

def cleaned_reading_minutes(doc):
    raw = doc.get('estimated_reading_time')
    if not raw:
```

```

        return 0.0 #accounts for potential problems
    raw = str(raw) #making sure that all is a string first

    digits = ''.join(ch for ch in raw if ch.isdigit()) # keep only digits by checking
    if digits == '':
        return 0.0

    return float(digits)

print('successfully cleaned')

```

successfully cleaned

```

In [13]: #now find all stories with read time above 60 minutes

twentyplus = collection.find({},
    {'story_name': 1,
     'estimated_reading_time': 1,
     'average_rating': 1,
     '_id': 0})

for doc in twentyplus:
    minutes = cleaned_reading_minutes(doc) #tells to use the cleaned minutes instead
    if minutes > 60: #checks if minutes are greater than 20
        print(doc['story_name'], minutes, 'minutes')

```

The Body Farm 74.0 minutes
 Ted the Caver 86.0 minutes
 A Story From the BlackGuard 139.0 minutes
 Tales From the Gas Station 75.0 minutes
 Ubloo 112.0 minutes
 Restoration 69.0 minutes
 Human Nature 68.0 minutes
 All of Our Mistakes Are Never Forgotten 76.0 minutes
 My Father and I Created a Device That Warps Reality 69.0 minutes
 Oppression 84.0 minutes
 The Hollows and the Hills 69.0 minutes
 The Hidden Webpage 64.0 minutes
 Tales From an Ex-Convict 201.0 minutes
 Cursed Sands 73.0 minutes
 The Remover 105.0 minutes
 The Stalker 67.0 minutes
 The Duxbury Chronicles 155.0 minutes
 Humper Monkey's Ghost Story 99.0 minutes
 The Church in the Woods 96.0 minutes
 Jeff The Killer: Scars of Corruption 86.0 minutes
 Nightmare 70.0 minutes
 Butcherface 133.0 minutes
 And Mira 149.0 minutes
 Amber Alert 98.0 minutes
 The 12 Rules 86.0 minutes
 Happy Boy 63.0 minutes
 Tribunal 75.0 minutes
 A Summer of Wolves 87.0 minutes

```

In [14]: #now find all of the outliers (the highly rated stories that are also short)

```

```

outliers = collection.find({},
    {'story_name': 1, 'estimated_reading_time': 1, 'average_rating': 1, '_id': 0})

for doc in outliers:
    minutes = cleaned_reading_minutes(doc)
    if doc['average_rating'] > 9 and minutes < 5:
        print(doc['story_name'], doc['average_rating'], minutes)

```

Distorted Warning Signals 9.12 3.0
 Sadie 9.08 1.0
 Lightning 9.08 2.0
 An Egg 9.08 4.0
 Survival Guide 9.08 4.0
 Julia Legare 9.06 3.0
 Candle Cove 9.04 4.0
 Persuaded 9.04 4.0
 An Open Letter To My Daughter's Killer 9.03 2.0
 The Only Sensible Ritual Pasta 9.03 3.0
 Sarah O'Bannon 9.03 1.0
 The Three Day Dark 9.02 3.0
 Never Again 9.02 3.0
 My Fear Of Water 9.02 2.0

In [15]: *#use Pandas to find the five longest stories*

```

longeststories = list(collection.find({},
    {'story_name': 1, 'estimated_reading_time': 1, 'average_rating': 1, '_id': 0}))

for d in longeststories:
    d['minutes'] = cleaned_reading_minutes(d)

top5 = sorted(longeststories, key=lambda x: x['minutes'], reverse=True)[:5] #now th

#print all of the stories
for doc in top5:
    print(doc['story_name'], doc['minutes'], ('minutes'), doc['average_rating'])

```

Tales From an Ex-Convict 201.0 minutes 8.79
 The Duxbury Chronicles 155.0 minutes 8.45
 And Mira 149.0 minutes 7.51
 A Story From the BlackGuard 139.0 minutes 9.04
 Butcherface 133.0 minutes 7.86

In [16]: *#find the average reading time for all stories*

```

times = []

for doc in collection.find({}, {'estimated_reading_time': 1, '_id': 0}):
    times.append(cleaned_reading_minutes(doc))

print('average reading time: ', sum(times) / len(times), 'minutes') #calculate the

```

average reading time: 9.647578347578348 minutes

In [17]: *#find the average length and rating for each section of categories*

```

pipeline = [

```



```

    {'$unwind': '$categories'},
    {'$group': {
        '_id': '$categories',
        'avg_rating': {'$avg': '$average_rating'},
        'avg_length': {'$avg': '$estimated_reading_time'},
        'count': {'$sum': 1}
    }},
    {'$sort': {'count': -1}}
]

```

```

category_stats = list(collection.aggregate(pipeline))[:5]
category_stats

```

```

Out[17]: [{'_id': ' Beings and Entities\n',
  'avg_rating': 7.448120805369127,
  'avg_length': None,
  'count': 298},
 {'_id': ' Strange and Unexplained\n',
  'avg_rating': 7.239890909090909,
  'avg_length': None,
  'count': 275},
 {'_id': ' Beings and Entities ',
  'avg_rating': 7.873600000000001,
  'avg_length': None,
  'count': 250},
 {'_id': ' Strange and Unexplained ',
  'avg_rating': 7.644356846473029,
  'avg_length': None,
  'count': 241},
 {'_id': ' Madness, Paranoia, and Mental Illness\n',
  'avg_rating': 7.34546875,
  'avg_length': None,
  'count': 128}]

```

In [18]: *#find the distribution of stories by their tags (show the first fifteen)*

```

pipeline = [{'$unwind': '$tags'},
    {'$group': {'_id': '$tags', 'count': {'$sum': 1}}},
    {'$sort': {'count': -1}}]

distributionofthetags = list(collection.aggregate(pipeline))[:15]
distributionofthetags

```



```

Out[18]: [{'_id': ' Video Narratives OK\n', 'count': 106},
          {'_id': ' anonymously authored\n', 'count': 85},
          {'_id': ' based on a true story, based on true events\n', 'count': 75},
          {'_id': ' parodies\n', 'count': 33},
          {'_id': ' Michael Whitehouse\n', 'count': 20},
          {'_id': ' Video Narratives OK\nPlease wait...\n', 'count': 17},
          {'_id': ' based on a true story, based on true events, Video Narratives OK\n',
            'count': 16},
          {'_id': ' Slenderman\n', 'count': 16},
          {'_id': ' aliens\n', 'count': 14},
          {'_id': ' undead, zombies\n', 'count': 14},
          {'_id': ' Crappypasta Success Stories\n', 'count': 11},
          {'_id': ' StupidDialUp\n', 'count': 10},
          {'_id': ' games, haunted games, Haunted Games Writing Contest\n',
            'count': 10},
          {'_id': ' conspiracies\n', 'count': 9},
          {'_id': ' David Feuling\n', 'count': 9}]

```

```

In [29]: #find all stories with the tag 'ghost'

```

```

allstories_with_ghost = collection.find({'tags': {'$regex': 'ghost', '$options': 'i',
          {'story_name': 1,
            'tags': 1,
            'average_rating': 1,
            '_id': 0}})

for doc in allstories_with_ghost [:15]:
    print(doc)

```

```
{'story_name': 'I Drive for Cerber', 'average_rating': 9.09, 'tags': ' apps, cars, c
reatures, driving, ghosts, internet, jobs, Mofucious, monsters, mythical creatures,
occupational, occupations, paranormal, spirits, taxi cabs, taxis, technology, Uber
\n'}
{'story_name': 'Forget Me', 'average_rating': 9.09, 'tags': ' beings, emotional, ent
ities, feelspastas, ghost stories, ghosts, spirits, Video Narratives OK, WatchThe-Af
termath\n'}
{'story_name': 'Dandelions', 'average_rating': 9.08, 'tags': ' afterlife, deaths, di
sappearances, dying, feelspastas, flowers, ghosts, happy endings, love, love storie
s, missing persons, murders, out-of-body experiences, paranormal, Stephanie Scissom
\n'}
{'story_name': 'A Peculiar Kind of Madness', 'average_rating': 9.06, 'tags': ' being
s, bell, entities, ghosts, Matt Dymerski, mysteries, strange\n'}
{'story_name': 'It Has No Face', 'average_rating': 9.04, 'tags': ' Ghost_Eye_Tree
\n'}
{'story_name': 'Sarah O'Bannon', 'average_rating': 9.03, 'tags': ' anonymously autho
red, burials, buried alive, cemeteries, deaths, ghosts, graveyards, spirits, undead,
zombies\nPlease wait...\n'}
{'story_name': 'Eastgate', 'average_rating': 8.99, 'tags': ' emotional, feelspastas,
ghost stories, ghosts, Ryan Peacock, spirits\n'}
{'story_name': 'I've Had the Same Dream for 26 Years', 'average_rating': 8.97, 'tag
s': ' Allan Loe, dreams, ghosts, medium, nightmares, occult, psychics, sleep, spirit
s\n'}
{'story_name': 'Darkstar', 'average_rating': 8.97, 'tags': ' creatures, crimes, cryp
tids, deaths, entities, fae, fairies, ghosts, murders, Myrretah, rapes, Shawnti Ther
rien, soulless, souls, spirits, strange, unexplained, werewolves\n'}
{'story_name': 'Raymond's Will', 'average_rating': 8.95, 'tags': ' beings, Benjamin
James, deaths, entities, ghost stories, ghosts, murders, paranormal, revenge, spirit
s\n'}
{'story_name': 'The Family Spirit', 'average_rating': 8.87, 'tags': ' ghost, ghosts,
haunting, spirit, Tobias Wade\nPlease wait...\n'}
{'story_name': 'Stories For My Daughter', 'average_rating': 8.87, 'tags': ' bedtime
stories, children, deaths, feelspastas, ghost stories, Halloween, Halloween stories,
HrhDaf, kids, memories, murders, poison, poisonings, spirits, twist endings\n'}
{'story_name': 'Love', 'average_rating': 8.84, 'tags': ' angels, anonymously authore
d, beings, deaths, entities, feelspastas, ghost stories, ghosts, guardian angels, lo
ve, spirits, strange, unexplained\n'}
{'story_name': 'The Most Terrifying 911 Call I've Ever Received', 'average_rating':
8.79, 'tags': ' 911, 911 calls, Cole Long, deaths, dispatchers, ghosts, jobs, murder
s, occupational, occupations, paranormal, police, unexplained\n'}
{'story_name': 'Room to Spare', 'average_rating': 8.78, 'tags': ' attractions, death
s, ghost stories, ghost tours, Halloween, Halloween attractions, Halloween stories,
haunted attractions, murders, Rhonnie Fordham, serial killers, tours, twist endings
\n'}
```

In [32]: *#find all stories with a rating below 3.5*

```
horrid_stories = collection.find({'average_rating': {'$lt': 3.5}},
    {'story_name': 1,
     'average_rating': 1,
     'estimated_reading_time': 1,
     '_id': 0})

for doc in horrid_stories:
    print(doc)
```

```
{'story_name': 'Your Life Is A Movie', 'average_rating': 3.43, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'Run Across the Desert', 'average_rating': 3.41, 'estimated_reading_time': '5 minutes'}
{'story_name': 'Grame', 'average_rating': 3.31, 'estimated_reading_time': '< 1 minute'}
{'story_name': 'The Blue Meteor', 'average_rating': 3.23, 'estimated_reading_time': '11 minutes'}
{'story_name': 'Karma', 'average_rating': 3.07, 'estimated_reading_time': '3 minutes'}
{'story_name': 'Katrina', 'average_rating': 3, 'estimated_reading_time': '7 minutes'}
{'story_name': 'A Haunting Entropy', 'average_rating': 2.7, 'estimated_reading_time': '13 minutes'}
{'story_name': 'The Town Of Blanche', 'average_rating': 2.63, 'estimated_reading_time': '2 minutes'}
{'story_name': 'Horrors of the Commonwealth', 'average_rating': 1.78, 'estimated_reading_time': '3 minutes'}
{'story_name': 'The N.Y. Subway Deaths', 'average_rating': 0.86, 'estimated_reading_time': '6 minutes'}
```

In [8]: *#now, convert the collection to a Pandas dataframe (remove body for length purposes)*

```
df_all = pd.DataFrame(list(collection.find({}, {'body': 0})))
df_all.head()
```

Out[8]:

	_id	story_name	average_rating	tags	estimated_readi
--	-----	------------	----------------	------	-----------------

0	6a1fa1933d3f30759140cbab	If You're Armed and at the Glenmont Metro, Ple...	9.25	drug trials, drugs, experimentation, experime...	12
---	--------------------------	---	------	--	----

1	6a1fa1933d3f30759140cbac	A Shattered Life	9.13	disease, entities, Matt Dymerski, monsters, p...	14
---	--------------------------	------------------	------	--	----

2	6a1fa1933d3f30759140cbad	My Father Punished Me When I Talked to Ghosts	9.12	abductions, blindness, child abuse, childhood...	8
---	--------------------------	---	------	--	---

3	6a1fa1933d3f30759140cbae	Distorted Warning Signals	9.12	Ashley Rose Wellman\n	3
---	--------------------------	---------------------------	------	-----------------------	---

4	6a1fa1933d3f30759140cbaf	He Who Wanders	9.12	Simon Simonian\n	22
---	--------------------------	----------------	------	------------------	----

In [36]: *#find the last five records of the dataframe*

```
five_final = collection.find({}, {'story_name': 1, 'average_rating': 1, 'estimated_
for doc in five_final:
    print(doc)
```

```
{'story_name': 'The N.Y. Subway Deaths', 'average_rating': 0.86, 'estimated_reading_
time': '6 minutes'}
{'story_name': 'Horrors of the Commonwealth', 'average_rating': 1.78, 'estimated_rea
ding_time': '3 minutes'}
{'story_name': 'The Town Of Blanche', 'average_rating': 2.63, 'estimated_reading_tim
e': '2 minutes'}
{'story_name': 'A Haunting Entropy', 'average_rating': 2.7, 'estimated_reading_tim
e': '13 minutes'}
{'story_name': 'Katrina', 'average_rating': 3, 'estimated_reading_time': '7 minute
s'}
```

In [39]: *#find all of the columns in the dataframe*

```
columnar_find = collection.find_one()
print(list(columnar_find.keys()))
```

```
['_id', 'story_name', 'average_rating', 'tags', 'body', 'estimated_reading_time', 'p
ublish_date', 'categories']
```

MongoDB Conclusions

In [42]: `print("MongoDB served more as a structural counterpoint than just a technical addit`

MongoDB served more as a structural counterpoint than just a technical addition to the project. While K-Means and KNN needed the dataset to be squished into numerical vectors, MongoDB let the data puff back up into its natural, narrative form. Each story—including all its metadata, categories, and text—was saved as a separate document. This approach treated stories not as parts of a flat table, but as individual pieces in a bigger connected system, each carrying its own logic yet fitting into larger trends too.

Adding a numeric `reading_minutes` field was not simply about making things easier; it was essential for the function of the database. Those original '___ minutes' strings in the `estimated_reading_time` column had meaning when read, but were detrimental for data analysis. Turning those strings into numbers meant the database's tools could actually work with that time information, just like the clustering and classification methods did. With this new field, the database could highlight previously hidden patterns: how long stories were, if length influenced ratings, and what categories often went together in the overall collection.

The queries—like sorting, filtering, and grouping—showed the data's underlying structure. Categories acted as tight clusters while tags formed looser links. Stories had spots in these frameworks based on length, rating, and focus. With MongoDB's aggregation pipeline, we saw these connections without dumbing down the data into just numbers. So MongoDB played a crucial role: it was flexible for keeping the story's complexity and had enough structure to show governing patterns.

Together, the MongoDB piece finishes the project design. It gave the needed details to see the dataset as a system, by virtue of its collection feature. This allowed the data to be viewed from statistical, structural, and narrative views. MongoDB integrates everything. It ties the different analyses together neatly, allowing one to grasp the dataset as an interconnected system with its own logic, and not merely as a bunch of separate tales.

In []: